

PhD Pathway — Marius Ngaboyamahina

A consolidated, source-backed strategy document. The candidate is profiled in third person; the roadmap, application strategy, and positioning statements are written in first person, exactly as they will be used.

1. Executive summary

Marius Ngaboyamahina is a senior software engineer with over 10 years of production experience, currently holding three concurrent roles from his base in Accra, Ghana: Senior Software Developer Lead (Consultant) at Credit Jambo Ltd, leading technical delivery of the Jambo360 digital-financial platform (Nov 2025–present); Senior Software Engineer Consultant at Hence Technologies Ltd (London-based legal-tech, June 2025–present), where he works in Palantir Foundry on data integration and workflow automation for global legal clients; and Digital Factory Manager at UMWALIMU SACCO in Kigali (June 2023–present), where he leads core-banking modernization, IFRS-9 automation, and a portfolio of fintech integrations. He holds an MSc in Information Systems from the Kobe Institute of Computing (Japan, 2019), a BSc in Computer Engineering from the University of Rwanda (2013), and was an ABE Initiative scholar of the Japanese government. He has one IEEE publication on sentiment analysis (ICBDA 2019) and is CCNA-certified. His profile is unusually strong on systems delivery, microservices, and multi-language engineering (PHP, JavaScript/TypeScript, SQL, Node.js, Java, Python, C#/.NET 7, React) — but comparatively light on recent academic publications and advanced ML theory.

Given that pattern, the strongest PhD pathway is **applied, systems-flavored AI research**, anchored in three priority directions: (1) AI for software engineering and developer tooling, (2) ML systems and data-centric AI infrastructure, and (3) trustworthy AI for fintech and other high-stakes regulated domains. These directions convert his existing strengths — large production codebases, financial-domain reliability constraints, and end-to-end delivery — into research credibility, while sidestepping the pure-theory ML competition where his profile would be weakest.

The recommended target portfolio is structured in three tiers: **Tier 1** salaried European doctoral positions where supervisor fit is decisive (U Luxembourg SnT/TruX, UvA, TU Delft AISE, Edinburgh CDTs); **Tier 2** highly competitive structured programs and fellowships (ETH AI Center, EPFL EDIC, IMPRS-IS, OIST, MSCA Doctoral Networks); and **Tier 3** national/government scholarships compatible with his Rwandan citizenship and Ghana base (MEXT 2027, DAAD, Commonwealth 2027/28, Knight-Hennessy, Gates Cambridge, plus Africa-anchored options like ARISE/ATPS). The full ranked list of 37 opportunities — with

deadlines, eligibility, fit scores, and recommended actions — lives in the companion spreadsheet [phd_funding_consolidated.xlsx](#).

The next 12 months are decisive. To convert his industry profile into a competitive academic application, he needs to produce two to three concrete research artifacts (a fintech-focused bug-report-to-code linker, a RAG evaluation framework for regulated-domain assistants, and a low-resource Kinyarwanda fintech NLP benchmark), publish at least one workshop or short paper, and run a structured outreach campaign to 8–12 priority supervisors before mid-2026 application cycles.

2. Consolidated profile analysis (third person)

2.1 Academic background

The candidate holds a BSc in Computer Engineering (University of Rwanda) and an MSc in Information Systems (Kobe Institute of Computing, Japan), giving him a credible two-degree CS-adjacent foundation. The MSc program in Japan is recognized internationally. It demonstrates international mobility, an ability to work in a second-language environment (English-medium with N4 Japanese), and exposure to systems-and-architecture coursework. This is a sufficient academic baseline for most applied AI/CS PhD programs, though it is not a research-master's track (e.g., MPhil, MSc by research, or a thesis-heavy ETH/EPFL master's), which is the strongest possible feeder for theory-intensive labs.

2.2 Technical strengths

More than ten years of senior engineering experience across regulated fintech (UMWALIMU SACCO, Credit Jambo / Jambo360), legal-tech (Hence Technologies / Palantir Foundry), live-streaming microservices (Cardinal Peak, USA), and IoT (Rexvirt, Tokyo) is the candidate's most differentiated asset. Concretely, this includes: backend and microservices design in PHP/Yii, Node.js/TypeScript, Java/Spring Boot, and C#/.NET 7; SQL/PostgreSQL data modeling at production scale; integrations with payment rails (MTN Escrow, RSwitch / RNDPS, PostBridge ATM) and regulator-grade reporting; IFRS-9 expected-credit-loss automation; Kafka and event-driven systems on AWS (MSK, MQ, SQS, EKS); CI/CD and containerization (Docker, Kubernetes); ETL for the Ishyiga data warehouse; and currently, hands-on Palantir Foundry work for a London-based legal-tech platform. For PhD admissions in AI for software engineering, ML systems, and trustworthy AI, this is exactly the kind of "real-world system at scale" experience that thesis committees value but most early-career applicants lack.

2.3 Research-relevant skills

Several capabilities transfer directly into research work: designing and evaluating systems end-to-end, building and operating data pipelines, running experiments in applied environments, translating between stakeholders and engineering, and producing working artifacts on deadline. These are particularly valuable in ML-systems research, software-engineering research, applied NLP, and human-centered AI, where empirical, build-and-measure work is the norm.

2.4 Publication and research exposure

The candidate has one peer-reviewed IEEE publication on sentiment analysis (2019) and an MSc that included systems and architecture coursework. This is enough to satisfy "evidence of research aptitude" for most applied programs, but it is light by the standards of top-10 US/European theory programs, where competitive applicants typically have 2–4 conference publications by the time they apply. The pragmatic implication is that he should target programs and supervisors who weigh applied/industry impact heavily, while strengthening his publication record over the next 12 months with at least one workshop paper.

2.5 Leadership and industry strengths

For a PhD applicant, the candidate's leadership profile is unusually strong: technical lead roles at three concurrent employers, direct architecture collaboration with a CTO at Credit Jambo, formal mentoring (African Leadership International / The Room's Tech Career Catalyst, WiredIn Academy, supervising a Carnegie Mellon embedded-systems practicum student at Cardinal Peak), and delivery responsibility on regulator-touching banking systems. The current Hence Technologies engagement adds Palantir Foundry experience — directly relevant to data integration, workflow assurance, and ontology-driven analytics in regulated domains. Programs that explicitly value applied impact, industry collaboration, or trustworthy/responsible AI applied to real domains will treat this combination as a major positive — particularly groups embedded in interdisciplinary centers (SnT Luxembourg, AISE TU Delft, Edinburgh CDTs, ETH AI Center).

2.6 Weaknesses and risks

Three honest weaknesses to plan around:

1. **Light recent publication record.** Only one publication, and it is from 2019. Mitigation: target one workshop paper or short paper from his project portfolio before December 2026 applications.
2. **No visible thesis-style research pipeline on the resume.** Mitigation: build the three project artifacts described in §6, all designed to be submittable to relevant venues.
3. **Less evidence of advanced math/theory depth** than applicants targeting foundational ML or formal verification programs. Mitigation: do not target those programs. Aim for

applied ML, ML systems, AI for SE, and trustworthy AI labs where his systems profile is an asset, not a liability.

2.7 Geographic and citizenship context

The candidate is a Rwandan citizen currently resident in Accra, Ghana, with prior in-country experience in Japan (Kobe MSc + Tokyo internship at Rexvirt). This is an advantage rather than a constraint: he is eligible for Commonwealth scholarships (Rwanda is a Commonwealth member), ATPS/ARISE Africa-anchored programs, and Mastercard Foundation Scholars Program tracks. He is also eligible for MEXT and most European scholarships, which are nationality-blind, and his prior **ABE Initiative scholarship** (Japanese government program for African students) plus **N4 Japanese** are concrete signals of fit for MEXT and OIST. The base in Ghana provides time zone and visa flexibility for European outreach and travel. The companion spreadsheet flags eligibility per program.

3. Refined research focus

Synthesizing the three source documents (Perplexity, ChatGPT, Gemini) and the candidate's profile, his sharpest viable research focus is:

Applied AI for the engineering and operation of trustworthy software systems in regulated, data-intensive domains — with a particular emphasis on developer tooling, data-centric ML infrastructure, and evaluation/assurance for retrieval-augmented and language-model-based systems used in fintech and core-banking workflows.

This framing is deliberately narrow enough to be credible (it ties to specific labs and venues), but broad enough to be portable across the three priority directions in §4. It allows him to write one strong research statement and selectively re-skin it per supervisor.

Three sub-themes anchor this focus:

- **Code intelligence with regulated-domain grounding.** Bug report understanding, code-change impact analysis, and automated repair, evaluated on fintech/banking codebases rather than the typical open-source datasets.
- **Data-centric infrastructure for assistive ML.** Data quality, lineage, retrieval, and evaluation infrastructure for LLM and RAG systems used in professional/regulated workflows.
- **Trustworthy AI assurance for high-stakes deployment.** Evaluation, calibration, monitoring, and human-in-the-loop assurance for AI used in financial decisioning and operations.

4. Ranked PhD direction shortlist

Direction 1 — AI for software engineering and developer tooling

- **What it is.** Research on AI systems (LLMs, retrieval, program analysis, ML-on-code) that help developers write, debug, review, test, and maintain software at scale.
- **Why does it rank first?** Cleanest path from senior engineer to researcher. The candidate's debugging, architecture, code review, and mentoring history transfer almost directly. Venues include ICSE, FSE, ASE, and MSR — all of which accept strong empirical work without requiring heavy theory.
- **Concrete topics.** Bug-report-to-code linking with retrieval-augmented LLMs; AI-assisted refactoring under regulatory constraints; evaluation frameworks for AI coding assistants in regulated codebases; defect prediction with calibrated uncertainty.
- **Top labs/supervisors:** Tegawendé F. Bissyandé ([SnT TruX, U Luxembourg](#)); Maliheh Izadi ([AISE Lab, TU Delft](#)); Denys Poshyvanyk (William & Mary); Westley Weimer (U Michigan); Reid Holmes (UBC); Ajitha Rajan ([Edinburgh Informatics](#)).
- **Publication expectation.** Medium. ICSE/FSE/ASE/MSR.
- **Transition realism.** Very high.

Direction 2 — ML systems and data-centric AI infrastructure

- **What it is.** Research on data, infrastructure, training, and serving systems that determine real-world AI performance and reliability — closer to MLSys, OSDI, and SIGMOD than to NeurIPS theory.
- **Why does it rank high?** Uses the candidate's strongest skills: production systems, data engineering, integration, and operational reliability. The publication bar emphasizes well-engineered measurement and benchmarks rather than novel mathematics.
- **Concrete topics.** Data quality and lineage for RAG; efficient inference in constrained environments; MLOps for high-stakes domains; evaluation infrastructure for enterprise AI; observability for deployed LLM systems.
- **Top labs/supervisors:** ETH Systems & ML groups via [ETH AI Center](#); EPFL ML4Systems via [EDIC](#); [Edinburgh CDT in Machine Learning Systems](#); Mahesh Marina (Edinburgh).
- **Publication expectation.** Medium. MLSys, NSDI, SIGMOD, VLDB.
- **Transition realism.** High.

Direction 3 — Trustworthy AI for fintech and high-stakes domains

- **What it is.** Research on robust, calibrated, interpretable, auditable, and human-supervised AI for domains where errors are costly.

- **Why does it rank high?** Aligned with the candidate's IFRS-9 and SACCO operational background. Many funded European programs (Edinburgh CDT in Responsible NLP, KU Leuven IAmAI, ETH AI Center) explicitly target this.
- **Concrete topics.** Calibrated risk models for credit decisions; assurance pipelines for RAG over regulatory documents; distribution-shift monitoring in fintech ML; ontology-grounded LLM workflows in regulated domains (a direct extension of his Palantir Foundry work at Hence); human-in-the-loop oversight for LLM-based decision support.
- **Top labs/supervisors:** Sara Magliacane ([UvA AMLab](#)); Ivan Titov (Edinburgh); [KU Leuven IAmAI](#) cohort; [ETH AI Center](#) supervisors in trustworthy ML.
- **Publication expectation.** Medium-high.
- **Transition realism.** Medium-high.

Direction 4 (secondary) — Applied NLP for enterprise/regulated workflows

A natural extension of the candidate's existing IEEE NLP publication, particularly attractive at [Edinburgh CDT in Responsible NLP](#), [ILCC Edinburgh](#), and OIST (Yamada). Strong overlap with Direction 1 in terms of methods.

Direction 5 (secondary) — Low-resource and African-language NLP / AI for development

A differentiated angle that leverages the candidate's Rwandan/East African context. ATPS/ARISE, Masakhane-affiliated supervisors, and African AI initiatives can be a credible Tier-3 backstop. The risk is that funding is thinner and supervisors are fewer; treated as a complement, not a primary direction.

Direction 6 (deprioritized) — Foundational ML theory and formal methods

Not recommended as the primary route. The candidate's profile lacks the recent math/theory depth most competitive theory programs require, and the opportunity cost of rebuilding that base in 12 months is high relative to the gains in Directions 1–3.

5. 6–12-month preparation roadmap (first person)

I will execute the next 12 months in five overlapping tracks. The cadence is designed around the European doctoral cycle: most Tier 1 vacancies repost between September 2026 and February 2027 for Fall 2027 starts.

Months 1–2 (May–June 2026): Foundation and signal

- I will publish a clean GitHub portfolio and a one-page personal research site, both linked from a refreshed Google Scholar profile.
- I will pick **three** target supervisors from §8 and read 3–5 of each of their most recent papers; for each, I will write a one-page critique plus one concrete extension I could pursue.
- I will refresh modern ML/NLP fundamentals via [CS224N \(Stanford\)](#) or the [Hugging Face NLP course](#), 6–8 hours/week, with a focused Anki deck.
- I will draft a single 2-page Statement of Research Interest, designed to be re-skinned per program.

Months 3–4 (July–August 2026): Project execution starts

- I will start **Project A** (bug-report-to-code linker for fintech codebases — see §6).
- I will register for and attend at least one online research event (e.g., MSR/ICSE virtual sessions, [Deep Learning Indaba](#)) and post a thoughtful summary on LinkedIn or my site.
- I will run my first round of supervisor outreach (3 emails) using the templates in §7, anchored on the critiques I wrote in Months 1–2.

Months 5–6 (September–October 2026): Apply for the early cycle

- I will submit the [MEXT 2027 embassy application](#) (Ghana embassy, opens April–May 2026, closes mid-year).
- I will submit [Knight-Hennessy](#) (early October 2026 deadline) and [Gates Cambridge US round](#) (16 October 2026), plus the international round in December.
- I will submit [DAAD](#) by 15 October 2026 for October 2027 entry.
- I will continue Project A and start drafting **Project B** (RAG evaluation framework — §6).
- I will do a second round of supervisor outreach (4 more emails).

Months 7–9 (November 2026–January 2027): Heavy application window

- I will apply to [EPFL EDIC](#) (1 December 2026 deadline) and [IMPRS-IS](#) (mid-November 2026).
- I will apply to [Edinburgh CDT in Responsible NLP](#) and [Edinburgh CDT in ML Systems](#) when their 2027 cohorts reopen (~January 2027 deadline based on prior cycles).
- I will submit any open Luxembourg SnT/TruX vacancy and UvA Human-Centered AI for Programming postings on [werkenbij.uva.nl](#).
- I will target submission of Project A as a workshop or short paper (e.g., MSR Mining Software Repositories, ICSE NIER, or an ML4SE workshop).

Months 10–12 (February–April 2027): Late cycle and decision

- I will apply to [OIST](#) (1 September is the next standard deadline; OIST runs a separate workshop on 15 November), and to [MSCA Doctoral Networks](#) host calls (the 2026 call opens 28 May 2026, deadline 24 November 2026).
- I will submit a Commonwealth PhD application when the 2027/28 cycle opens (~September 2026, based on the 2026/27 closing date of 14 October 2025).
- I will close on Project B and start **Project C** (low-resource Kinyarwanda fintech NLP benchmark — §6) as a Year-1 thesis seed.
- I will conduct interviews, request supervisor calls, and make a final decision by April–May 2027.

Continuous tracks (every month)

- 1 supervisor email or follow-up per week.
 - 3–5 hours/week of supervised reading from the [Connected Papers](#) and [Semantic Scholar](#) maps I am building (see §10).
 - 1 GitHub commit per week to one of the three projects.
 - One LinkedIn or research-site post every two weeks reflecting on a paper I read or a result from one of my projects.
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6. Research-grade project plan

I will produce three artifacts in the next 12 months. Each is designed to be (a) credible to a thesis committee, (b) submittable to a real venue, and (c) tied to my fintech background so it is not generic.

Project A — Fintech bug-report-to-code linker

One-line description. A retrieval-augmented system that maps natural-language bug reports from production financial systems to the most likely code locations and prior fixes, with calibrated confidence.

Why it matters. Bug-report-to-code linking is a long-standing problem in software engineering (ICSE/FSE/MSR). The novelty I bring is evaluating it on regulated-domain codebases I have direct working access to (anonymized SACCO/IFRS-9 modules and Jambo360 microservices) where defect cost is high and ground-truth fix history is dense — a dataset profile that is essentially absent from the public ML-for-SE literature.

Method. Sentence-Transformer or code-aware embedding for issues and code; hybrid BM25+dense retrieval; LLM-based re-ranking with citation requirement; calibrated confidence

via temperature scaling. Baselines: BM25, plain dense retrieval, and a small fine-tuned re-ranker.

Deliverables. Public anonymized benchmark (subset of the codebase with synthetic but realistic issues), reproducible code on GitHub, and a 4–6 page short paper targeting MSR Mining Software Repositories or ICSE NIER.

Timeline. Months 3–8.

Target venue. [MSR 2027](#) or ICSE NIER 2027.

Project B — RAG evaluation framework for regulated-domain assistants

One-line description. An open-source evaluation harness that scores RAG/LLM systems on factual grounding, citation faithfulness, and refusal calibration when answering questions over a curated regulatory document corpus (e.g., IFRS-9 standards, central-bank circulars).

Why it matters. Most RAG evaluations use generic open-domain QA. Regulated domains care about citation faithfulness and conservative refusal more than answer fluency. This is a clear open gap in trustworthy-AI evaluation.

Method. Curated regulatory document set; multi-axis scoring (faithfulness, citation precision/recall, refusal calibration, answerability); harness compatible with [LangChain](#) and [LlamaIndex](#) backends; comparison across 3–5 open and proprietary LLMs.

Deliverables. Open-source repo, leaderboard-style report, and a workshop paper for [TrustNLP](#) or a NeurIPS workshop on evaluation.

Timeline. Months 5–10.

Target venue. TrustNLP 2027 or a NeurIPS 2027 workshop.

Project C — Kinyarwanda fintech NLP benchmark (thesis seed)

One-line description. A small, high-quality benchmark for low-resource Kinyarwanda NLP focused on fintech tasks: customer-service intent classification, regulatory text understanding, and code-mixed Kinyarwanda/English bug report classification.

Why it matters. It is differentiated, geography-relevant, and aligned with the [Masakhane](#) movement. It functions as a credible Year-1 thesis seed for any of the three priority directions and is particularly attractive to ATPS/ARISE and African-AI–anchored funders.

Method. Data collection from public regulator publications and synthetic dialogue generation with native-speaker validation; baseline models (mBERT, AfroXLMR, fine-tuned multilingual LLM); honest evaluation including failure analysis on dialect and code-mixing.

Deliverables. Open dataset (CC-licensed where possible), baselines on GitHub, and a paper for AfricaNLP or [Findings of ACL/EMNLP](#).

Timeline. Months 8–12 (initial release); becomes a Year-1 thesis chapter post-admission.

Target venue. AfricaNLP workshop or Findings of EMNLP 2027.

7. Networking and target-lab strategy

7.1 Prioritization

I will run a tight, high-quality outreach campaign rather than a high-volume one. The plan is **12 priority supervisors** in three tiers (4-4-4), each contacted with a tailored email referencing a specific recent paper.

7.2 Outreach cadence

- Weeks 1–4: 3 emails (top tier).
- Weeks 5–8: 3 emails (mid tier).
- Weeks 9–12: 3 emails (third tier).
- Months 4–6: 3 follow-ups + 3 new emails as new vacancies appear.

7.3 Cold email template (first person)

Subject: Senior fintech engineer interested in your work on [specific topic] — PhD inquiry

Dear Professor [Last Name],

I am Marius Ngaboyamahina, a senior software engineer with over ten years of experience. I currently lead the technical team for Credit Jambo's Jambo360 digital-financial platform, consult on Palantir Foundry workflows for Hence Technologies (London legal-tech), and run the Digital Factory at UMWALIMU SACCO in Rwanda. I am based in Accra, Ghana, and hold an MSc in Information Systems from the Kobe Institute of Computing as an ABE Initiative scholar of the Japanese government.

I read your recent paper on [exact title] and was particularly struck by [one specific technical observation, not generic praise]. It connects directly to a problem I have hit repeatedly in production — for example, in IFRS-9 automation at UMWALIMU SACCO or in our Foundry-based legal workflows at Hence — [one-sentence concrete example]. I have started a small prototype that [one-sentence description of Project A or B], and I would like to push this into a doctoral research direction with the rigor your group brings.

I am preparing PhD applications for Fall 2027 and your group is one of my top targets. Two questions, if I may:

1. Are you considering taking new doctoral students for the 2027 intake, and if so, are funded vacancies expected to be posted in [approx. window]?
2. Would you be open to a 20-minute conversation in the next month to discuss whether my background and interests are a fit for your agenda?

Attached: 1-page CV; link to my GitHub portfolio: [link]; link to my draft research statement: [link].

Thank you for your time.

Best regards, Marius Ngaboyamahina [email] | [LinkedIn] | [Scholar] | [GitHub]

7.4 Conference and event targets

- [Deep Learning Indaba 2026](#) — high-leverage African AI network.
- [ICSE 2027](#), [MSR 2027](#), [FSE 2027](#), virtual or hybrid attendance.
- [EurNLP](#) and ACL/EMNLP virtual.
- [WiNLP](#) and [LatinX in AI](#) for community building.

7.5 Tracking system

I will maintain a Notion or Airtable database with one row per supervisor: name, lab, last-email date, last-response date, paper read, response status, fit score (1–10), and next-action date. The same database will track applications.

8. Supervisor and lab targeting framework

This is my Tier 1 list. The companion spreadsheet expands the broader funding/program set; this section is about people and groups I plan to contact directly.

Supervisor / Lab	Institution	Why fit (third person)	Source
Tegawendé F. Bissyandé — TruX	U Luxembourg, SnT	ERC Fellow; AI for program repair, mining software repositories. Africa-friendly. The candidate's fintech codebase experience is a direct match for TruX's work on bug fixing and program repair.	TruX, SnT
Maliheh Izadi — AISE Lab	TU Delft	LLMs for software engineering; growing group with strong industry collaboration. Fits Direction 1.	AISE / TU Delft
Sebastian Proksch / Arie van Deursen	TU Delft	DevOps, software engineering at scale. Strong match for Project A.	TU Delft SERG
Sara Magliacane — AMLab	UvA	Causal AI; trustworthy ML; supervises in the Human-Centered AI for Programming track. Direction 3.	AMLab
Ivan Titov	Edinburgh ILCC	NLP, interpretability, structured prediction. Direction 4 + 3.	ILCC
Ajitha Rajan	Edinburgh Informatics	Software testing, ML for SE. Direction 1.	Edinburgh Informatics
Mahesh Marina	Edinburgh Informatics	Networked / ML systems. Direction 2.	CDT in ML Systems

Supervisor / Lab	Institution	Why fit (third person)	Source
Reid Holmes — Software Practices Lab	UBC	Empirical SE, AI for SE, developer tools. Direction 1.	SPL UBC
Denys Poshyvanyk	William & Mary	Deep learning for SE, ACM Fellow. Direction 1.	W&M CS
Westley Weimer	U Michigan	PL/SE, automated program repair. Direction 1.	U-M CSE
Makoto Yamada — MLDS	OIST	ML, NLP, low-resource methods. Direction 4–5.	OIST MLDS
Brooke / Mendes group — Human-Centered AI for Programming	UvA Informatics	Interactive developer tooling, HCI+AI. Direction 1.	UvA Informatics

For each supervisor I will record: 3 most recent papers read, 1 critique + extension idea, 1 connection point from his industry experience, and the specific funding mechanism that would support a position (group budget, ERC, MSCA, CDT, university scholarship).

9. Funding and application strategy

The detailed 37-row funding matrix is in the companion spreadsheet [phd_funding_consolidated.xlsx](#) — that is the primary artifact for this section. The spreadsheet ranks every opportunity 1–37 with fit score (1–10), eligibility, deadline, and a recommended action. The strategy below is the narrative layer.

9.1 Tiered portfolio

- **Tier 1 — salaried European doctoral positions (priority).** These are full employment contracts (≈€2,000–4,000/month) where supervisor's fit decides the outcome, and there is no separate scholarship competition. Targets: U Luxembourg SnT/TruX, UvA, TU Delft AISE, MSCA Doctoral Networks. These are spreadsheet rows ranked 1–6.

- **Tier 2 — competitive structured programs and fellowships.** Cohort-based, prestigious, requires a strong research statement and supervisor pre-contact. Targets: ETH AI Center, EPFL EDIC, IMPRS-IS, OIST, Edinburgh CDTs, KU Leuven IAmAI.
- **Tier 3 — national and government scholarships compatible with my profile.** [MEXT 2027](#) (Ghana embassy track), [DAAD](#), [Commonwealth 2027/28](#), [Knight-Hennessy](#), [Gates Cambridge](#), ARISE/ATPS, Mastercard Foundation Scholars Program.

9.2 Verified deadlines (May 2026 cutoff)

The spreadsheet contains the full list with citations. The most consequential ones for my next 12 months:

- **Vanier CGS Canada — discontinued in 2024**, excluded from the plan ([Government of Canada announcement](#)).
- **MEXT 2027** — embassy applications open April–May 2026 (Ghana eligible) ([Study in Japan](#)).
- **DAAD** — opens May 2026, closes 15 October 2026 for October 2027 entry ([DAAD](#)).
- **Commonwealth PhD 2026/27** closed 14 October 2025; next call ~September 2026 for 2027/28 ([CSC](#)).
- **MSCA Doctoral Networks 2026** call opens 28 May 2026, deadline 24 November 2026 ([European Commission](#)).
- **OIST** — next deadline 1 September 2026 for Spring 2027; admissions workshop 15 November 2026 ([OIST](#)).
- **EPFL EDIC** — 1 December 2026 deadline for September 2027 ([EPFL EDIC](#)).
- **IMPRS-IS** — November 2025 cycle closed; expect ~mid-November 2026 for 2027 ([IMPRS-IS](#)).
- **Edinburgh CDT in Responsible NLP** — 7 January 2026 closed; expect ~September–October 2026 reopen ([Responsible NLP CDT](#)).
- **Edinburgh CDT in ML Systems** — expect ~October–November 2026 reopen for September 2027 ([CDT in ML Systems](#)).
- **ETH AI Center Doctoral Fellowships** — 5–8 fellowships annually, cycle runs in late autumn ([ETH AI Center](#)).
- **Knight-Hennessy** — early October 2026 deadline ([Knight-Hennessy](#)).
- **Gates Cambridge** — 16 October 2026 (US round), December 2026/January 2027 (international) ([Gates Cambridge](#)).
- **ARISE / ATPS** — annual cycle; 2026 closed 31 March 2026; African-anchored, strong fit ([ATPS](#)).
- **AISG PhD Singapore** — August 2026 intake closed; August 2027 cycle opens ~Q1 2027 ([AISG](#)).
- **Australian National Industry PhD** — closed 20 March 2026; next ~Q1 2027 ([NIPhD](#)).

9.3 Application priorities (first person)

If I had to pick five applications that I will treat as non-negotiable in the next 12 months, they are:

1. The first U Luxembourg SnT/TruX vacancy that opens in my area (Bissyandé).
2. UvA Human-Centered AI for Programming (any 2027 reposting).
3. MEXT 2027 (Ghana embassy, opens April–May 2026).
4. EPFL EDIC (1 December 2026).
5. Edinburgh CDT — whichever (Responsible NLP or ML Systems) reposts first for the 2027 cohort.

Everything else in the spreadsheet is layered on top of these five.

9.4 Risk flags

The spreadsheet has a "Main risk" column. The most consequential cross-cutting risks I will manage:

- **Fellowship vs. position confusion.** Some Tier 1 European posts are employment contracts requiring specific vacancy postings; I cannot apply when no vacancy exists. I will set Google Alerts and check the institution job boards weekly.
- **Eligibility drift.** Some scholarships exclude applicants who already hold a master's (e.g., Chevening is positioned as a master's-level award; flagged in the spreadsheet as low priority for my level).
- **Verification gaps.** Where the source documents listed claims I could not confirm, the spreadsheet flags them in the "Status" column as "verify before applying."

10. AI-powered research workflow

I will use AI tools as a research assistant, not as a replacement for primary sources. The cadence below is what I will run weekly.

Day	Task	Tool(s)	Output
Mon	Topic scoping for the week	Perplexity , ChatGPT , Gemini Deep Research	A short listing of 5–10 candidate papers and 2 open questions

Day	Task	Tool(s)	Output
Tue	Paper discovery	Semantic Scholar , Connected Papers , Google Scholar alerts	A reading list of 5 papers
Wed	Deep reading	Elicit , Scite , Consensus	Notes in Zotero with tags
Thu	Drafting (writing day)	ChatGPT/Claude as editor, Grammarly for prose, Overleaf for LaTeX	A draft section of the research statement or a project log
Fri	Code/experiment day	GitHub Copilot , Cursor, ChatGPT for code review	Working notebook or PR pushed
Sat	Outreach + admin	Email templates from §7	1–2 supervisor emails
Sun	Reflection	Notion or Obsidian	A 1-page weekly review: what I learned, what I will change next week

Tool roles, briefly:

- **Perplexity / Gemini Deep Research / ChatGPT** — search, scoping, "explain this concept" prompts.
- **Elicit / Consensus / Scite** — literature triage; I never cite from these tools without reading the underlying paper.
- **Connected Papers / ResearchRabbit / Semantic Scholar** — citation graphs for finding adjacent work.
- **Zotero + Better BibTeX** — single source of truth for references.
- **Cursor / Copilot** — code velocity for the three projects.
- **Notion or Obsidian** — research journal, supervisor tracker, application tracker.

Guardrails I will keep.

- I will never let an LLM write the substance of my research statement; LLMs edit, I write.
- I will always verify funding/scholarship claims against the official program page, never an aggregator.

- I will keep a "decisions and assumptions" log so I can explain why I chose any particular framing or paper.
-

11. Ready-to-use prompts

The eight prompt categories below are the ones I will reuse most often. Each is written so I can paste it into Perplexity, ChatGPT, or Gemini Deep Research with minor edits.

11.1 Supervisor and lab discovery

"Act as a PhD admissions strategist. Find current professors and labs working on [topic, e.g., 'retrieval-augmented LLMs for software engineering'] who are actively taking PhD students for Fall 2027 in [Europe / North America / Asia]. For each, give: full name, institution, lab URL, 2–3 representative papers from 2024–2026, current funding/vacancy signals, and Africa-friendly track record. Cite official lab pages, not aggregator sites."

11.2 Literature mapping

"Build a literature map of [topic] from 2022–2026. Cluster the 30 most cited papers into 4–6 sub-themes. For each sub-theme, list the canonical paper, the most recent advance, the dominant evaluation benchmark, and 2 open problems. Output as a markdown table."

11.3 Research-gap identification

"Given this set of papers [paste 5 abstracts], identify 3 specific, fundable research gaps a PhD student could pursue in 3–4 years. For each gap, propose a concrete first paper, the dataset/benchmark required, the most likely reviewer objections, and the strongest single result that would make it accept-worthy."

11.4 Research statement drafting

"I am writing a 2-page research statement for a PhD in [topic]. My background is: [paste 1-paragraph bio]. My target supervisor is [name] working on [paper title]. Draft an outline with: (1) motivating problem, (2) why my background is differentiated, (3) two concrete research questions, (4) preliminary work I will reference, (5) a one-paragraph alignment with the supervisor's recent work. Keep it specific and evidence-based. Do not write the prose yet — only the outline."

11.5 Cold email drafting

"Polish this cold email to a professor for grammar and tone, but do not change the technical content. Make it warmer in greeting, more specific in the middle, and crisper in the ask. Limit to 220 words." (Then paste my draft of the §7.3 template.)

11.6 Funding eligibility check

"I am a Rwandan citizen resident in Ghana with a master's degree from Japan, applying for a PhD starting Fall 2027. For [scholarship name], extract from the official page: eligibility (citizenship, residency, prior degree), exact deadline, what the funding covers, the application steps, and any non-obvious risks. Cite the official URL only."

11.7 Project planning

"I have 12 months to deliver a publishable artifact in [topic]. Given my background [paste bio], propose a feasible scope, the smallest novel contribution that is still defensible, the dataset/benchmark, the baseline I must beat, and a month-by-month milestone plan. Flag the top 3 risks and how to mitigate them."

11.8 Mock interview / committee prep

"Act as a tough but fair PhD interview committee. Ask me 8 questions in escalating difficulty about my proposal on [topic]. After each of my answers, give one strength and one specific improvement. End with a verdict: 'fund', 'borderline', 'not yet,' and the single most important thing I should change."

12. Final positioning statements (first person)

Three crisp, reusable statements I will adapt across emails, applications, and conversations.

12.1 Long form (≈140 words; for research statements, motivation letters)

I am a senior software engineer with over ten years of experience, currently splitting my time across three regulated-domain engagements: leading the technical team for the Jambo360 digital-financial platform at Credit Jambo, building Palantir Foundry workflows for a London-based legal-tech firm at Hence Technologies, and running the Digital Factory at UMWALIMU SACCO where I shipped IFRS-9 expected-credit-loss automation, online lending, and the integrations with Rwanda's national payment system. I came to AI through systems: my MSc in Information Systems at the Kobe Institute of Computing (ABE Initiative scholar of the Japanese

government) and my IEEE publication on sentiment analysis grounded my interest in applied NLP, and a decade of industry work has made it clear where AI actually breaks in real systems — bug triage at scale, code maintenance under regulatory constraints, and the assurance gap around language-model-based assistants in financial and legal workflows. My PhD goal is to do empirical, systems-flavored research on AI for software engineering, ML systems, and trustworthy AI — building artifacts that work in regulated, data-intensive environments rather than only on benchmarks. I am specifically targeting groups that combine engineering rigor with high-stakes deployment as a first-class concern.

12.2 Medium form (≈55 words; for cold emails, networking)

I am a Rwandan senior software engineer with ten years across fintech (Credit Jambo, UMWALIMU SACCO) and legal-tech (Hence Technologies / Palantir Foundry), an MSc from Japan as an ABE Initiative scholar, and one IEEE NLP publication. I am applying for a Fall 2027 PhD in AI for software engineering, ML systems, or trustworthy AI, and my goal is research grounded in regulated, real-world systems — not benchmark-only work.

12.3 Short form (≈22 words; for elevator/intro)

Senior fintech and legal-tech engineer with ten years in regulated systems across Rwanda, Ghana, and Japan — applying for a 2027 PhD in applied AI for trustworthy software systems.

Sources

This document synthesizes three uploaded source files (PhD_pathway_ChatGPT.md, PhD_pathway_Perplexity.md, PhD_pathway_Gemini.md) and a funding spreadsheet/CSV that the candidate provided, deduplicated and reconciled against official program pages where they conflicted. All funding claims, deadlines, and program statuses in this document and in [phd_funding_consolidated.xlsx](#) are cited inline from the official funder or university page (not from aggregator sites). Where the source documents made claims I could not verify on an official page, the spreadsheet's "Status" column flags them as "verify before applying," and the document does not rely on them.